

LABTASK 1:

SOURCE CODE:

```
#include <iostream>
using namespace std;
class Person{
protected:
    int age;
    string name;
public:
    Person(char n, int a){
        cout << "I AM IN BASE CLASS"<<endl;
        name = n;
        age = a;
    }
    ~Person(){
        cout << "BASE CLASS DESTRUCTOR"<<endl;
    };
    void get_data(){
        cout << "ENTER NAME: ";
        cin >> name;
        cout << "ENTER AGE:" ;
        cin >> age;
    }
    void show_data(){
        cout << "NAME: " << name << endl;
        cout << "AGE:" << age <<endl;
    }
};
```

```

class Student: public Person{
private:
    float fee;
    int roll_number;
public:
    Student(float f, int r, int a, char n): Person(n, a){
    }
    void set_Student_data(int r)
    {
        get_data();
        roll_number = r;
    }
    void calculate_fee(float f){
        fee = f;
        fee = fee+1000;
    }
    void show_student_data(){
        show_data();
        cout <<"ROLL NUMBER: " <<roll_number<<endl;
        cout << "FEE: " <<fee<<endl;
    }
};

```

```

int main(){
    Student s1(24000.90, 1, 25, 'W');

    s1.set_Student_data(1);
    s1.calculate_fee(2200);
    s1.show_student_data();
    return 0;
}

```

OUTPUT:

```

I AM IN BASE CLASS
ENTER NAME: w
ENTER AGE:18
NAME: w
AGE:18
ROLL NUMBER: 1
FEE: 3200
BASE CLASS DESTRUCTOR
PS C:\Sem 2\OOP LABS\lab11>

```

LABTASK 2:

SOURCE CODE:

```
#include <iostream>
using namespace std;

class person{
protected:
    string name;
    int age;
public:
    void setPersonDetail(string n, int a){
        name = n;
        age = a;
    }
    void displayPersonDetail(){
        cout<<"NAME: "<<name<<endl;
        cout<<"AGE: "<<age<<endl;
    }
};
```

```
class AcademicRecord: virtual public person{
protected:
    float GPA;
    string major;
public:
    void setAcademicDetails(float gpa, string m){
        GPA = gpa;
        major = m;
    }
    void displayAcademicDetails(){
        cout<<"GPA: "<<GPA<<endl;
        cout<<"MAJOR: "<<major<<endl;
    }
};
```

```
class scholarship : virtual public person, public AcademicRecord{
private:
    int scholarshipAmount;
public:
    void evaluateScholarship(){
        if (GPA>3.5 && age>25){
            scholarshipAmount = 50000;
        }
        else{
            scholarshipAmount=0;
        }
    }
    void displayScholarshipDetails(){
        cout<<"SCHOLARSHIP AWARDED: "<<scholarshipAmount<<endl;
    }
};
```

```

int main(){
    scholarship s1;
    s1.setPersonDetail("WALEEJA", 18);
    s1.setAcademicDetails(3.3, "AI");
    s1.evaluateScholarship();

    s1.displayPersonDetail();
    s1.displayAcademicDetails();
    s1.displayScholarshipDetails();
    cout<<endl;

    scholarship s2;
    s2.setPersonDetail("ALI", 29);
    s2.setAcademicDetails(3.89, "CS");
    s2.evaluateScholarship();

    s2.displayPersonDetail();
    s2.displayAcademicDetails();
    s2.displayScholarshipDetails();

    return 0;
}

```

OUTPUT:

```

PS C:\Sem 2\OOP LABS\lab11> g++ 07.cpp
NAME: WALEEJA
AGE: 18
GPA: 3.3
MAJOR: AI
SCHOLARSHIP AWARDED: 0

NAME: ALI
AGE: 29
GPA: 3.89
MAJOR: CS
SCHOLARSHIP AWARDED: 50000
PS C:\Sem 2\OOP LABS\lab11>

```

